

The Helix–Hilbert–Polya Bridge: Source Completeness and Shifted Readout Closure

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Abstract

This note records the mathematical core currently formalized in Lean and evaluated by the numerical model. The Hilbert–Polya side is the R1–R9 infrastructure chain. For non-principal channels it supplies source completeness: every nontrivial zero is realized by a three-dimensional source fiber event carrying the $\pi/3$ coordinate translation. The crossing projection is now formalized directly: the source coordinate is shifted to the $\pi/3$ midpoint $\pi/6$, the one-dimensional logarithmic readout is $1/2$, and the height is retained. Thus the shifted-coordinate discharge is unconditional in Lean for all non-principal channels. The zeta/L1 channel is handled separately through the eta-regularized zeta bridge with the same shifted-coordinate readout.

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1 Objects and Notation

Let χ be a Dirichlet character modulo q and write

$$L(s, \chi)$$

for Mathlib's Dirichlet L -function. For non-principal χ , define the strip zero set

$$\text{NTZ}(\chi) := \{\rho \in \mathbb{C} : 0 < \text{Re } \rho < 1, L(\rho, \chi) = 0\}.$$

The per-character GRH predicate formalized in Lean is

$$\text{GRH}(\chi) := \forall \rho \in \text{NTZ}(\chi), \quad \text{Re } \rho = \frac{1}{2}.$$

The all non-principal channel closure predicate used in the current Lean theorem is

$$\text{GRH}_{\neq 1} := \forall q, \forall \chi \pmod{q}, \quad \chi \neq 1 \implies \text{GRH}(\chi).$$

The principal zeta/L1 channel is treated separately through eta regularization. For zeta, let

$$\text{NTZ}_\zeta := \{\rho \in \mathbb{C} : 0 < \text{Re } \rho < 1, \zeta(\rho) = 0\}.$$

2 The $\pi/3$ Helix Geometry

2.1 Canonical Scale

The numerical model and the formal gauge use the canonical $\pi/3$ chart

$$A = \frac{\pi}{6}, \quad ds = \frac{\pi}{3}, \quad C = 2A \, ds = \left(\frac{\pi}{3}\right)^2.$$

The exact area-law placement is

$$R_n^2 = Cn, \quad R_n = \frac{\pi}{3}\sqrt{n}.$$

At the source level the data are the $\pi/3$ coordinates, the channel weights coming from residue counting, and the harmonic phase on the helix. The fiber is derived from that source. No one-dimensional continuation formula is used to create it. The standard symbols L , ζ , and Hurwitz tails below occur because parts of the derived fiber identify with those representations after the geometric fiber has already been placed. Thus the three-dimensional source fiber atom at height t , for $s = \frac{1}{2} + it$, is

$$a_n(s) = w(n) \frac{1}{R_n} \exp\{-it \text{Log}(R_n^2)\} = w(n)(Cn)^{-s}.$$

Lemma 2.1 (Area-law closed form). *For any channel weights $w(n)$ for which the corresponding Dirichlet series is defined,*

$$F_w(s) := \sum_{n \geq 1} w(n)(Cn)^{-s} = C^{-s} \sum_{n \geq 1} w(n)n^{-s}.$$

Proof. Use $R_n^2 = Cn$ and $s = \frac{1}{2} + it$:

$$R_n^{-1} e^{-it \operatorname{Log}(R_n^2)} = (R_n^2)^{-1/2} (R_n^2)^{-it} = (R_n^2)^{-s} = (Cn)^{-s}.$$

Summing gives the result. □

2.2 Non-principal Character Channels

For a non-principal Dirichlet character χ modulo q , the channel weights are

$$w(n) = \chi(n).$$

Therefore the source channel sum is

$$F_\chi(s) = \sum_{n \geq 1} \chi(n)(Cn)^{-s}.$$

In the projected one-dimensional notation this same harmonic sum is

$$F_\chi(s) = C^{-s} L(s, \chi).$$

Residue-wise, with zero residues omitted, the same placed channel may be reindexed as

$$F_\chi(s) = C^{-s} q^{-s} \sum_{r \bmod q} \chi(r) \zeta\left(s, \frac{r}{q}\right).$$

For χ_3 ,

$$\chi_3(1) = 1, \quad \chi_3(2) = -1, \quad \chi_3(0) = 0,$$

and hence

$$F_{\chi_3}(s) = C^{-s} 3^{-s} \left[\zeta\left(s, \frac{1}{3}\right) - \zeta\left(s, \frac{2}{3}\right) \right].$$

2.3 The Principal/Zeta Channel

For the zeta/L1 channel the fiber is eta-regularized:

$$w(n) = (-1)^{n+1}.$$

The source channel sum is

$$F_{L1}(s) = \sum_{n \geq 1} (-1)^{n+1} (Cn)^{-s}.$$

In the projected one-dimensional notation the eta factor is

$$\eta(s) = \sum_{n \geq 1} (-1)^{n+1} n^{-s} = (1 - 2^{1-s}) \zeta(s).$$

Thus the L1 fiber is

$$F_{L1}(s) = C^{-s}\eta(s) = C^{-s}(1 - 2^{1-s})\zeta(s).$$

Equivalently, in residue form modulo 2,

$$F_{L1}(s) = C^{-s}2^{-s} \left[\zeta\left(s, \frac{1}{2}\right) - \zeta(s, 1) \right].$$

On the critical line $\operatorname{Re} s = \frac{1}{2}$, the eta factor is nonzero:

$$1 - 2^{1-s} \neq 0,$$

because $2^{1-s} = 1$ would imply $2^{1-\operatorname{Re} s} = 1$, hence $\operatorname{Re} s = 1$.

2.4 FTA and Euler Product Generation

The source helix is prime-generated. Its integer fibers are generated from prime coordinates by unique factorization, and the Euler product is the convergence-half-plane readout of that multiplicative geometry.

Proposition 2.2 (FTA helix character). *Fix a prime-angle assignment $\theta : \mathbb{N} \rightarrow \mathbb{R}$ and define the FTA winding angle*

$$\Theta_\theta(n) = \sum_{p^e \parallel n} e \theta(p).$$

For $m, n \geq 1$,

$$\Theta_\theta(mn) = \Theta_\theta(m) + \Theta_\theta(n).$$

Consequently the unit-circle winding

$$\omega_\theta(n) = \exp(i\Theta_\theta(n)),$$

is multiplicative:

$$\omega_\theta(mn) = \omega_\theta(m)\omega_\theta(n).$$

For a Dirichlet channel χ , the channel helix character

$$\Psi_{\chi, \theta}(n) = \chi(n)\omega_\theta(n)$$

is multiplicative:

$$\Psi_{\chi, \theta}(mn) = \Psi_{\chi, \theta}(m)\Psi_{\chi, \theta}(n) \quad (m, n \geq 1).$$

Proof. By unique factorization,

$$v_p(mn) = v_p(m) + v_p(n)$$

for every prime p . Summing $e \theta(p)$ over the finite factorization support gives the displayed additivity of Θ_θ . Exponentiating converts addition of angles into multiplication on the unit circle. Since Dirichlet characters are multiplicative, the product $\chi(n)\omega_\theta(n)$ is multiplicative as well.

Lean anchors:

```
HelixLogFree.windAngle_mul
HelixLogFree.wind_mul
HelixMult.helixChar_mul
HelixMult.helixChar_one
```

□

Proposition 2.3 (Log bridge and Euler product readout). *The logarithm enters only at the external analytic bridge. With bridge angle*

$$\theta_\gamma(p) = \gamma \log p,$$

the FTA identity

$$\log n = \sum_{p^e \parallel n} e \log p$$

gives

$$\Theta_{\theta_\gamma}(n) = \gamma \log n, \quad \omega_{\theta_\gamma}(n) = n^{i\gamma}.$$

In the convergence half-plane $\operatorname{Re} s > 1$, the same multiplicativity is read as an Euler product:

$$\operatorname{HelixSource}_\chi^C(s) = \sum_{n \geq 1} \chi(n) (Cn)^{-s} = C^{-s} \prod_p (1 - \chi(p)p^{-s})^{-1}.$$

Taking the logarithmic derivative gives the prime-power trace generated by the Euler product:

$$\mathcal{T}_\chi^C(s) = C^{-s} \sum_{n \geq 1} \Lambda(n) \chi(n) n^{-s} = C^{-s} \left(-\frac{L'}{L}(s, \chi) \right).$$

The winded prime-field bridge is

$$\sum_{n \geq 1} \chi(n) \Lambda(n) \omega_{\theta_\gamma}(n) n^{-s} = -\frac{L'}{L}(s - i\gamma, \chi), \quad \operatorname{Re} s > 1.$$

Where the product representation converges, it also gives the Euler-product edge fact

$$\operatorname{Re} s \geq 1 \implies L(s, \chi) \neq 0 \quad (\chi \neq 1).$$

Proof. The identity

$$\log n = \sum_{p^e \parallel n} e \log p$$

is the logarithmic image of unique factorization. Substituting $\theta_\gamma(p) = \gamma \log p$ into the FTA angle gives $\Theta_{\theta_\gamma}(n) = \gamma \log n$, hence $\omega_{\theta_\gamma}(n) = n^{i\gamma}$. In $\operatorname{Re} s > 1$, absolute convergence allows the completely multiplicative integer series to factor into its prime Euler product. Applying the logarithmic derivative to the Euler product gives the von Mangoldt prime-power trace. Multiplication by $\omega_{\theta_\gamma}(n) = n^{i\gamma}$ shifts the trace from s to $s - i\gamma$.

Lean anchors:

```
HelixWindBridge.real_log_eq_factorization_sum
HelixWindBridge.windAngle_glog
HelixWindBridge.wind_glog_eq_cpow
HelixMult.helixSource_eq_eulerProduct
HelixGauge.helixTrace_eq_gauge_mul_negLogDeriv
HelixWindBridge.windedVonMangoldt_eq_neg_logDeriv_shift
HelixMult.helix_no_zero_re_ge_one
```

□

Thus composites are not independent sources; they are generated by their prime coordinates through FTA. This is the structural distinction from a functional-equation-only construction. The product convergence is half-plane-limited; the FTA multiplicativity of the helix character itself is not height-limited.

For the principal L1 channel, the same prime generation is present with $\chi = 1$. The working fiber is eta-regularized,

$$F_{L1}(s) = C^{-s}(1 - 2^{1-s})\zeta(s),$$

and the zero readout uses the gauged zeta resolvent trace

$$C^{-s} \left(-\frac{\zeta'}{\zeta}(s) \right).$$

The eta factor handles the principal-channel pole; it does not replace prime generation.

2.5 Euler-Product Discriminator

The previous propositions are not used as a standalone zero-location argument. The logarithmic Euler product

$$\log \zeta(s) = \sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks}$$

is a convergence-half-plane identity. It is not evaluated at a strip zero. The role of the Euler product in the closure is instead to certify that the source fiber being projected is the prime-generated Dirichlet/zeta fiber, not only an object with a matching functional equation.

Definition 2.4 (Euler-product discriminator). *For a non-principal Dirichlet channel χ , the Euler-product discriminator*

$$\mathcal{E}_\chi$$

is the following package:

(E1) *the source fiber factors into the prime Euler product in $\text{Re } s > 1$:*

$$\text{HelixSource}_\chi^C(s) = C^{-s} \prod_p (1 - \chi(p)p^{-s})^{-1};$$

(E2) *the winded prime-power trace is the shifted logarithmic derivative:*

$$\sum_{n \geq 1} \chi(n) \Lambda(n) \omega_{\theta_\gamma}(n) n^{-s} = -\frac{L'}{L}(s - i\gamma, \chi), \quad \text{Re } s > 1;$$

(E3) *the Euler-product edge gives nonvanishing on $\text{Re } s \geq 1$:*

$$\text{Re } s \geq 1 \implies L(s, \chi) \neq 0.$$

In Lean this package is `EulerProductDiscriminator`. For Dirichlet channels it is constructed unconditionally by

$$\text{eulerProductDiscriminatorOfDirichlet}.$$

Proposition 2.5 (Discriminator entry point for the original-coordinate wrapper). *The original-coordinate Pythagorean wrapper consumes the Euler-product discriminator before it consumes the Pythagorean readout. In Lean the readout field has the form*

```

projected_readout_pythagorean :
  EulerProductDiscriminator chi ->
  FiberCaptureEvent chi rho ->
  SourceFiberEvent chi rho ->
  no_drift ->
  Proj chi rho ->
  normSq (w rho) = 1

```

The compatibility wrapper consumes this field in

```

SourceFiberEvent.midpoint_axis_of_HP
SpectralSide.w_unit_iff_half

```

before the midpoint conclusion

$$\operatorname{Re} \rho = \frac{1}{2}.$$

Proof. The wrapper first builds $\mathcal{E}_\chi$ from the Dirichlet Euler product and winded von Mangoldt trace. Only then does faithful projection provide the unit-circle/Pythagorean readout. Therefore a functional-equation-only model cannot pass through the same formal step unless it supplies the same discriminator package. This is the falsifier check: if the source object is replaced by a Davenport–Heilbronn-type readout with the same symmetry but no Euler-product source package, the theorem is not applicable at the readout step. $\square$

3 Closed-Form Accumulation and Resolvent Trace

The placed fiber accumulation and the resolvent-trace accumulation are distinct but linked closed forms.

3.1 Polya Operator Normal Form

For a non-principal channel, the placed source fiber is

$$F_\chi^C(s) = \sum_{n \geq 1} \chi(n) (Cn)^{-s} = C^{-s} L(s, \chi).$$

This is the channel amplitude accumulation: it is the harmonic sum of the source fiber terms in the $\pi/3$ area chart.

Lemma 3.1 (No radial drift in the accumulation normal form). *The helix source has no radial drift by construction. For each source atom*

$$a_n(t) = w(n) R_n^{-1} \exp\{-it \operatorname{Log}(R_n^2)\},$$

the radial size is independent of t :

$$|a_n(t)| = \frac{|w(n)|}{R_n} = \frac{|w(n)|}{(\pi/3)\sqrt{n}} \quad (C = (\pi/3)^2).$$

Proof. For real t and positive R_n^2 ,

$$|\exp\{-it \operatorname{Log}(R_n^2)\}| = 1.$$

The helix flow therefore changes only phase. The radial coordinate is fixed by the source placement $R_n^2 = Cn$, so the accumulated source is no-drift in the radial direction before projection. $\square$

This no-drift normal form is the invariant used by the Polya induction. Moving from one crossing to the next advances the phase ledger and harmonic threshold, not the radial placement. Hence R7 can be applied to the infinite ladder with the same $\pi/3$ source geometry at every step; combined with R1–R6 and R8–R9, this is the source-side path to `SourceComplete3D`.

Definition 3.2 (Admissible source harmonic domain). *For $s = \sigma + it$ and a finite set $F \subset \mathbf{N}$, the admissible finite source harmonic is*

$$H_{F,\chi}(s) = \sum_{n \in F} \chi(n) n^{-\sigma} U(t, n)^{-1}, \quad U(t, n) = \exp(it \log n).$$

For the initial segment $F = \{0, \dots, M\}$ this is the Lean object

$$\text{HelixFlowClosureLedger.flowPartialSum } \chi \text{ } s \text{ } M.$$

The admissible source domain is the increasing ladder obtained from these finite source harmonics by the threshold induction R6–R7, with the $\pi/3$ coordinate chart R8 applied at every crossing. No Green–Helmholtz energy norm is part of this R1–R9 source domain.

Proposition 3.3 (Source operator normal form). *The Polya source flow in R1–R3 is the diagonal unitary flow*

$$U(t, n) = \exp(it \log n), \quad \|U(t, n)\| = 1,$$

with real generator

$$H(n) = \log n, \quad \Im H(n) = 0.$$

Consequently a conserved nonzero source amplitude has zero radial drift.

Proof. R2 is the unitary one-parameter flow

$$\text{HelixFlow.isUnitaryOneParameterFlow.}$$

R3 is the computed Stone normal form

$$\text{HelixFlowGenerator.phasorFlow_eq_exp,} \quad \text{HelixFlowGenerator.gen_real,}$$

combined with the no-drift implication

$$\text{HelixSource.source_noDrift.}$$

This is the operator content used inside

$$\text{hilbertPolyaChainR1ToR9.}$$

□

Theorem 3.4 (The helix is its own continuation into the strip). *For a non-principal channel χ , define the finite source phasor chain*

$$S_M(s, \chi) = \sum_{0 \leq n \leq M} \chi(n) n^{-\sigma} U(t, n)^{-1}, \quad s = \sigma + it, \quad U(t, n) = \exp(it \log n).$$

This is the same three-dimensional helix fiber, projected through the area-law/log bridge. For every s with $\sigma > 0$ there is a constant $C_{\chi,s} > 0$ such that

$$|S_M(s, \chi) - L(s, \chi)| \leq C_{\chi,s} M^{-\sigma} \quad (M \geq 1).$$

Thus the helix source chain continues itself from the convergent half-plane into the critical strip. At a zero ρ with $0 < \operatorname{Re} \rho < 1$,

$$|S_M(\rho, \chi)| \leq C_{\chi, \rho} M^{-\operatorname{Re} \rho},$$

so the zero is read as a source fiber cancellation event. Moreover the rescaled chain exposes the height-free source ledger:

$$|S_M(\rho, \chi) M^\rho - (A_\chi(M) - L(0, \chi))| \leq C'_{\chi, \rho} M^{-1}, \quad A_\chi(M) = \sum_{0 \leq n \leq M} \chi(n).$$

Proof. The identity

$$S_M(s, \chi) = \text{HelixFlowClosureLedger.flowPartialSum } \chi \text{ } s \text{ } M$$

is the phasor-flow form of the source chain. The continuation estimate is

$$\text{HelixFlowClosureLedger.flow_rate_bound.}$$

The cancellation estimate at a zero is

$$\text{HelixFlowClosureLedger.flow_chain_vanishes_at_zero.}$$

The exact height-free ledger is

$$\text{HelixFlowClosureLedger.flow_closure_exact.}$$

The bundled Lean theorem

$$\text{helixOwnContinuationIntoStrip}$$

packages these three statements. This is not the divergent Euler product in the strip; it is the same harmonic source chain continued by the closure ledger. The zeta/L1 channel uses the parallel eta-regularized fiber and standing readout; on the midpoint readout the combined statement is the Lean theorem

$$\text{zetaL1HelixFiber_zero_iff_standingWave_node.}$$

It is built from:

$$\begin{aligned} & \text{HelixGauge.piThirdZetaFiber_zero_iff} \\ & \text{HelixStandingWave.zeta_zero_on_line_iff_standingWave_node} \end{aligned}$$

□

Proposition 3.5 (Pythagorean midpoint readout). *For a source fiber event, once faithful dimensional transport supplies the projected Pythagorean readout, the unit-circle coordinate forces*

$$\operatorname{normSq}(w(\rho)) = 1 \iff \operatorname{Re} \rho = \frac{1}{2}.$$

At the $\pi/3$ source scale this is the same midpoint statement

$$\operatorname{arcChart}(\operatorname{Re} \rho) = \frac{\pi}{6}.$$

Equivalently, if $z_2(\rho) \in S^1$ is the two-dimensional unit-circle projection of the crossing, written in the $\pi/3$ angular chart, then the one-dimensional readout is the lifted logarithm

$$\ell(\rho) = \frac{3}{\pi} \frac{1}{i} \text{Log}_{\mathcal{H}} z_2(\rho), \quad z_2(\rho) = \exp\left(i \frac{\pi}{3} \ell(\rho)\right),$$

where $\text{Log}_{\mathcal{H}}$ is the continuous helix branch selected by the crossing ledger. Thus the unit-circle projection is still a $\pi/3$ -coordinate object, and the 1D value is obtained by taking its logarithm and applying the same scale conversion.

Proof. The readout equivalence is

$$\text{SpectralSide.w_unit_iff_half},$$

equivalently `readout_pythagoras_iff_online` in the Mobius readout layer. The $\pi/3$ coordinate conversion is R8:

$$\text{arcChart_line}.$$

In the closure theorem this appears as

$$\text{SourceFiberEvent.midpoint_axis_of_HP}.$$

The lifted logarithmic readout notation is formalized by

$$\text{liftedCrossingCircleValue}, \quad \text{liftedCrossingLog}, \quad \text{piThirdLogReadout_midpoint}.$$

The value $1/2$ is therefore the unit-readout midpoint forced by the source sign-change/no-drift crossing after faithful projection, not an independent condition on the zero set. $\square$

Proposition 3.6 (Threshold ladder). *The first source crossing has amplitude cost $\pi/2$, and each subsequent harmonic threshold advances by π :*

$$\text{quantumLevelZ } 0 = \frac{\pi}{2}, \quad \text{quantumLevelZ}(k+1) = \text{quantumLevelZ}(k) + \pi.$$

At a threshold $E(t) = n\pi$, the harmonic count is exactly n .

Proof. These are the R6 clauses

$$\begin{aligned} &\text{quantumLevelZ_midpoint_straddle}, \\ &\text{quantumLevelZ_step}, \\ &\text{HelixProduction.harmonicCount_at_threshold}. \end{aligned}$$

The channel cancellation/rendezvous law is the remaining R6 clause, using

$$\text{vanishing_iff_rendezvous}, \quad \text{price_at_rendezvous}.$$

$\square$

Remark 3.7 (What does not enter R1–R9). *The Green–Helmholtz / von Neumann Gram package*

$$\text{GramVonNeumann}$$

and the finite Dirac package

$$\text{HelixDirac}$$

are not imported by `RequestProject/HilbertPolyaChain.lean` and are not used by the compiled declarations

`hilbertPolyaChainR1ToR9,`
`sourceComplete3D_of_HP,`
`SourceFiberEvent.midpoint_axis_of_HP.`

They are separate operator/energy models. The current Polya chain uses the phasor-flow unitarity, real generator/no-drift, threshold ladder, source cancellation, resolvent pole, $\pi/3$ chart, and Pythagorean readout path above.

Proposition 3.8 (Exact multiplicity in the trace). *Assume the boundary trace identity*

$$T(z) = -\frac{L'}{L} \left(\frac{1}{2} + iz, \chi \right).$$

If $\rho \in \text{NTZ}(\chi)$ has order m , then at $z = \text{poleParam}(\rho)$ the resolvent trace has a simple principal part with residue im . In the s -variable and $\pi/3$ gauge this is the same residue law

$$\text{Res}_{s=\rho} \mathcal{T}_\chi^C(s) = -mC^{-\rho}.$$

Proof. The exact principal-part transport is

`HelixLimit.multiplicityCapture_of_resolventTrace.`

It is built from the source log-derivative residue theorem

`HelixSource.LFunction_logDeriv_residue_eq_order`

through the coordinate $s = \frac{1}{2} + iz$. The multiplicity is therefore read from the Laurent residue of the resolvent trace, not from a numerical fit. $\square$

Theorem 3.9 (Source completeness from the Polya normal form). *For every non-principal χ , every fiber capture event $\rho \in \text{NTZ}(\chi)$ is realized by a source fiber event:*

$$\text{FiberCaptureEvent}_\chi(\rho) \implies \text{Nonempty}(\text{SourceFiberEvent}_\chi(\rho)).$$

Proof. The no-radial-drift normal form fixes the source placement $R_n^2 = Cn$ at every harmonic threshold. R7 then applies the ladder induction to all crossings without changing the source geometry. R5 supplies the source-chain cancellation and the resolvent pole; R8 supplies the $\pi/3$ coordinate chart and intrinsic ledger; and R9 identifies the Mathlib zero-set object. This is the Lean theorem

`sourceComplete3D_of_HP.`

$\square$

The resolvent trace is the logarithmic, prime-power-weighted accumulation. In the convergent half-plane,

$$\begin{aligned} \mathcal{T}_\chi^C(s) &:= \sum_{n \geq 1} \Lambda(n) \chi(n) (Cn)^{-s} \\ &= C^{-s} \sum_{n \geq 1} \Lambda(n) \chi(n) n^{-s} \\ &= C^{-s} \left(-\frac{L'}{L}(s, \chi) \right). \end{aligned}$$

Here Λ is the von Mangoldt weight. Thus the same source geometry has two closed-form accumulations: the fiber sum $C^{-s}L(s, \chi)$ and the resolvent trace $C^{-s}(-L'/L)(s, \chi)$. In Lean these are

$$\text{HelixGauge.helixSource_eq_gauge_mul_L}$$

and

$$\text{HelixGauge.helixTrace_eq_gauge_mul_negLogDeriv.}$$

At the canonical scale $C = (\pi/3)^2$, the continued trace is

$$\mathcal{T}_\chi^{\pi/3}(s) = \left(\left(\frac{\pi}{3} \right)^2 \right)^{-s} \left(-\frac{L'}{L}(s, \chi) \right),$$

which is the paper form of `HelixGauge.piThirdHelixTraceCont_eq`.

The literal logarithmic derivative of the placed fiber differs only by a regular gauge term:

$$-\frac{d}{ds} \log F_\chi^C(s) = \log C - \frac{L'}{L}(s, \chi).$$

The additional term $\log C$ has no pole. The multiplicative geometric gauge C^{-s} is never zero for $C > 0$. Therefore neither gauge changes the crossing locations or creates/removes trace poles; the multiplicative gauge only rescales the trace residue by a nonzero geometric factor.

This gives the local residue law at a capture event. If ρ is a zero of $L(s, \chi)$ of order m and

$$L(s, \chi) = (s - \rho)^m g(s), \quad g(\rho) \neq 0,$$

then near ρ

$$-\frac{L'}{L}(s, \chi) = -\frac{m}{s - \rho} - \frac{g'}{g}(s),$$

and hence

$$\mathcal{T}_\chi^C(s) = -\frac{mC^{-\rho}}{s - \rho} + \text{a term finite at } \rho.$$

So the trace pole has residue $-mC^{-\rho}$. The integer m is the spectral multiplicity; the factor $C^{-\rho}$ is the faithful $\pi/3$ geometric scaling. At a crossing, this is the residue consumed by the resolvent trace while the source fiber amplitude crests, the channel phase cancels, and the sign changes. The Polya threshold condition is therefore the simultaneous event

$$\begin{aligned} F_\chi^C(\rho) &= 0, & \text{Res}_{s=\rho} \mathcal{T}_\chi^C(s) &= -mC^{-\rho}, \\ E_\chi(\text{Im } \rho) &= A_k^\pm, & A_0^\pm &= \pm \frac{\pi}{2}, & A_{k+1}^\pm - A_k^\pm &= \pm \pi. \end{aligned}$$

Here E_χ is the source accumulation ledger used by the Polya chain. The jump from one level $A_k^\pm$ to the next is the threshold that starts the next harmonic on the spectral readout. Thus the closed-form resolvent residue, the crossing sign change, and the creation of the next harmonic are the same source event viewed in three ledgers: trace, standing wave, and harmonic ladder.

For primitive non-principal χ , the completed zero-side resolvent trace is the partial-fraction accumulation

$$\mathcal{R}_\chi(s) = \sum_{\rho \in \text{NTZ}(\chi)} m_\rho \left(\frac{1}{s - \rho} + \frac{1}{\rho} \right),$$

where m_ρ is the vanishing order of the completed L -function at ρ . The formal closed form is

$$\log \text{Deriv } \Lambda_\chi(s) = A + \mathcal{R}_\chi(s) \quad (s \notin \text{NTZ}(\chi)),$$

for a constant A . Splitting the completed function $\Lambda_\chi = \gamma_\chi L(s, \chi)$ gives, off the zeros in the strip,

$$-\frac{L'}{L}(s, \chi) = -A - \mathcal{R}_\chi(s) + \log \text{Deriv } \gamma_\chi(s).$$

Thus the prime-power accumulation, the zero-side resolvent trace, and the continued source trace are the same readout up to the constant and Archimedean gauge terms, which are regular away from their own gauge poles.

For the L1/zeta channel,

$$F_{L1}^C(s) = C^{-s}(1 - 2^{1-s})\zeta(s).$$

Off $\text{Re } s = 1$, the eta factor is nonzero, so a zero of the L1 fiber is a zero of ζ . The corresponding continued helix resolvent trace is

$$\mathcal{T}_{L1}^C(s) = C^{-s} \left(-\frac{\zeta'}{\zeta}(s) \right).$$

The eta factor regularizes the principal fiber; it is not inserted into the zero-pole readout, because the pole is the logarithmic derivative of the zeta factor whose zeros are being read. In Lean,

$$\text{HelixGauge.piThirdZetaTraceCont.eq}$$

is the $\pi/3$ specialization of this trace. The raw trace is

$$-\frac{\zeta'}{\zeta}(s),$$

and the $\pi/3$ zeta fiber zero has both a raw and a gauged resolvent-trace pole by

$$\text{HelixGauge.piThirdZetaFiber_zero_resolventTrace_pole,}$$

$$\text{HelixGauge.piThirdZetaFiber_zero_zetaTraceCont_pole.}$$

4 Standing Readout and Projection to One Dimension

This is the layer where the placed three-dimensional fiber is read in the lower-dimensional coordinate chart. The functions below are readouts of the source geometry and channel counts, with the coordinate phase correction kept explicit.

For $s = \frac{1}{2} + it$, the projected complex fiber is

$$F_\chi(t) = F_\chi\left(\frac{1}{2} + it\right).$$

The real standing readout has the form

$$Z_\chi(t) = \text{Re} \left(e^{i\Theta_\chi(t)} F_\chi\left(\frac{1}{2} + it\right) \right),$$

where

$$\Theta_\chi(t) = \arg \Gamma\left(\frac{1/2 + a}{2} + \frac{it}{2}\right) + \frac{t}{2} \log \frac{q}{\pi} + t \log C + \alpha_\chi.$$

Here $a \in \{0, 1\}$ is the parity and α_χ is the character/root-number gauge. For the L1/zeta eta-regularized channel one subtracts the eta gauge

$$\arg(1 - 2^{1-s}).$$

The term $t \log C$ is not a new oscillation. It is the coordinate correction from

$$\text{Log}(R_n^2) = \text{Log}(Cn) = \log n + \log C.$$

Omitting it rotates the real readout into the wrong coordinate chart.

4.1 Phase Conventions

All logarithms of positive real quantities are real logarithms. Thus

$$\text{Log}(R_n^2) = \log(Cn), \quad \log q, \quad \log C$$

use the real branch. The gamma phase in the standing readout is the continuous phase

$$\vartheta_\chi(t) = \text{Im} \log \Gamma\left(\frac{1/2 + a + it}{2}\right) + \frac{t}{2} \log \frac{q}{\pi},$$

with the branch chosen continuously along the vertical line $z(t) = (1/2 + a + it)/2$. The root-number phase is fixed by

$$\tau(\chi) = \sum_{r=1}^q \chi(r) e^{2\pi i r/q}, \quad \varepsilon_\chi = \frac{\tau(\chi)}{i^a \sqrt{q}}, \quad \alpha_\chi = -\frac{1}{2} \arg(\varepsilon_\chi).$$

For the principal zeta/L1 channel, $\alpha_\chi = 0$ and the two-channel eta readout uses the continuous phase

$$\phi_\eta(t) = \arg\left(1 - 2^{1-(1/2+it)}\right),$$

so that

$$\Theta_{L1}(t) = \vartheta_\zeta(t) + t \log C - \phi_\eta(t).$$

For non-principal χ ,

$$\Theta_\chi(t) = \vartheta_\chi(t) + t \log C + \alpha_\chi.$$

5 Harmonic Cost and the Quantum Ladder

The full source geometry is double ended. The midpoint is the origin of the fiber, and the reflected side is an anti-helix carrying the opposite orientation. The proof uses the positive oriented side of the helix for simplicity; the reflected anti-helix is not used in the formal proof closure. It is used in computational verification: the reflected side supplies the symmetric pre-origin geometry needed to build a high-precision tail before the counting ledger starts at the origin. Its cost law is the same with the sign reversed.

The crossing levels in the projected unit-circle readout are therefore

$$A_k = \left(k + \frac{1}{2}\right) \pi, \quad k \in \mathbb{Z}.$$

Equivalently, for $m \geq 0$ the two oriented ladders are

$$A_m^+ = \left(m + \frac{1}{2}\right) \pi, \quad A_m^- = -\left(m + \frac{1}{2}\right) \pi.$$

On the positive helix side used in the proof,

$$A_0 = \frac{\pi}{2}, \quad A_{k+1} = A_k + \pi.$$

Thus the first crossing is the only crossing on that side whose cost from the origin is a half quantum:

$$|A_0^+| = \frac{\pi}{2}.$$

Every additional crossing on that same side costs one full quantum:

$$A_{m+1}^+ - A_m^+ = \pi \quad (m \geq 0).$$

The reflected anti-helix obeys the same rule:

$$|A_0^-| = \frac{\pi}{2}, \quad |A_{m+1}^- - A_m^-| = \pi \quad (m \geq 0).$$

So the half-quantum occurs once at the first crossing on each oriented side of the double-ended helix; all subsequent crossings consume full π increments.

Conjecture 5.1 (Origin/Tate interface). *If this helix model has a Tate-thesis-style global/local interpretation, the natural place for that connection is the shared origin of the double-ended helix. The origin is where the two reflected orientations meet before the one-dimensional readout separates them into positive and negative crossing ladders.*

6 Hilbert–Polya Infrastructure: R1–R9

This section lists the infrastructure used before the shifted-coordinate crossing readout is applied. It does not assert an unshifted zero-location conclusion.

R1. Hilbert space. The source space is

$$\ell^2(\mathbb{N}; \mathbb{C}).$$

R2. Unitary phasor flow. The FTA/Euler phasor flow is

$$U(t)(n) = n^{it}, \quad |U(t)(n)| = 1,$$

with

$$U(s+t)(n) = U(s)(n)U(t)(n).$$

R3. Real generator and no-drift principle. The generator is

$$H(n) = \log n \in \mathbb{R}, \quad U(t)(n) = e^{itH(n)}.$$

The source no-drift lemma is: if

$$e^{\operatorname{Re} \lambda \tau} c = c \quad \text{for all } \tau, \quad c \neq 0,$$

then

$$\operatorname{Re} \lambda = 0.$$

R4. Helix-native trace/readout equals L . The finite phasor chain is

$$S_M(s, \chi) = \sum_{n \leq M} \chi(n) n^{-\operatorname{Re} s} U(\operatorname{Im} s)(n)^{-1}.$$

For $0 < \operatorname{Re} s$ and $\chi \neq 1$,

$$S_M(s, \chi) \rightarrow L(s, \chi)$$

with explicit rate:

$$\|S_M(s, \chi) - L(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

R5. **Zeros induce fiber capture events.** At a zero $L(s, \chi) = 0$ with $0 < \operatorname{Re} s$,

$$\|S_M(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

At every $\rho \in \operatorname{NTZ}(\chi)$,

$$-\frac{L'}{L}(s, \chi)$$

has no finite limit at ρ in the punctured neighborhood.

R6. **Spectral realization by crossing confluence and channel cancellation.** For any continuous strictly monotone accumulation E , if

$$E(a) \leq c \leq E(b), \quad a \leq b,$$

then there is a unique t such that

$$a \leq t, \quad E(t) = c.$$

At threshold $E(t) = n\pi$,

$$\left\lfloor \frac{E(t)}{\pi} \right\rfloor = n.$$

At a crossing threshold the source event is a confluence: the channel phase cancels, the amplitude reaches its crest, the resolvent trace consumes the local residue, the sign changes, and the next harmonic begins on the spectral readout. In the source normalization used by the Polya chain,

$$E(\operatorname{Im} \rho) = A_k^\pm, \quad A_0^\pm = \pm \frac{\pi}{2}, \quad A_{k+1}^\pm - A_k^\pm = \pm \pi.$$

So the first crossing on each oriented side costs a half quantum, and every subsequent threshold is one full π step. The channel cancellation law is:

$$\sum_{m \in F} s_m = 0 \iff P_F(s) = M_F(s),$$

where

$$P_F(s) = \sum_{m \in F} \max(s_m, 0), \quad M_F(s) = \sum_{m \in F} \max(-s_m, 0).$$

At cancellation,

$$\sum_{m \in F} |s_m| = 2P_F(s).$$

R7. **Exhaustion by ladder induction and rigidity.** If a property holds at A_0 and is preserved by $A_k \mapsto A_{k+1}$ and $A_k \mapsto A_{k-1}$, then it holds at every A_k . Moreover, any crossing sequence c_n satisfying

$$E(c_n) = n\pi$$

is forced to be the purchase-height sequence:

$$c_n = \operatorname{purchaseHeight}(n).$$

R8. **Coordinate faithfulness and Pythagorean readout.** The rechart

$$\text{arcChart}(x) = \frac{\pi}{3}x$$

has inverse

$$\text{arcChartInv}(u) = \frac{3}{\pi}u.$$

The critical midpoint is

$$\text{arcChart}(x) = \frac{\pi}{6} \iff x = \frac{1}{2}.$$

The value $1/2$ is therefore only the normalized one-dimensional readout of the source midpoint $\pi/6$. It is not an independently fitted spectral constant. At a source crossing, the standing readout changes sign; that sign change forces the crossing to occur at the midpoint of the source chart. The rule is source-side: projection is only responsible for transporting the event without losing it or introducing a new one. The midpoint conclusion routes through the unitary/Pythagorean readout. In natural coordinates the readout satisfies

$$(\text{readout}_{\sigma,\gamma})_{\text{Re}}^2 + (\text{readout}_{\sigma,\gamma})_{\text{Im}}^2 = 1 \iff \sigma = \frac{1}{2}.$$

Equivalently, for the spectral value $w(\rho) = 1 - 1/\rho$,

$$\|w(\rho)\|^2 = 1 \iff \text{Re } \rho = \frac{1}{2}.$$

The source conservation/no-drift input supplies the unitary source side of this readout.

R9. **Genuine zero-set object.** The object is Mathlib's L -function zero set:

$$\rho \in \text{NTZ}(\chi) \iff 0 < \text{Re } \rho < 1, \quad L(\rho, \chi) = 0.$$

7 Source Completeness and Projection Faithfulness

Definition 7.1 (Source crossing confluence). *For a channel χ and point $\rho = \sigma + i\gamma$, a source crossing confluence at height γ is a three-dimensional fiber event with the following simultaneous features:*

- (C1) *the channel phases cancel, so the fiber vanishes in the channel sum;*
- (C2) *the accumulated amplitude reaches the relevant π -level crest at one of the ladder levels $A_m^\pm$, with the first oriented crossing carrying the half-quantum cost $\pi/2$ and later crossings advancing by full π increments;*
- (C3) *the crossing consumes the threshold energy required to start the next harmonic on the spectral readout, so the outgoing side of the crossing begins the next harmonic;*
- (C4) *the real standing readout changes sign at the crossing;*
- (C5) *the source midpoint coordinate forced by the sign change and the real-axis/unitary readout requirement is $x = \pi/6$ in the $\pi/3$ chart, while the height γ is retained as the vertical coordinate.*

(C6) the same crossing location is projected from the three-dimensional helix to a two-dimensional unit-circle value $z_2(\rho)$ in the $\pi/3$ chart; the subsequent one-dimensional readout is obtained by taking the lifted logarithm of that unit-circle value.

Thus the source crossing itself determines the midpoint. Clause (C5) records that consequence; it is not an additional lower-dimensional placement assumption. After the normalization $x \mapsto 3x/\pi$, this source midpoint is the value $1/2$.

Under projection from the three-dimensional fiber to the unit-circle readout, radial and absolute phase information are collapsed. The retained datum is the crossing height γ , together with the unit-circle event

$$z_2(\rho) = \exp(iu_\rho), \quad u_\rho \in \frac{\pi}{3}\mathbb{R}.$$

The subsequent one-dimensional readout is the lifted logarithm

$$\text{Log}_{\mathcal{H}} z_2(\rho) = iu_\rho, \quad \ell(\rho) = \frac{3}{\pi}u_\rho = \frac{3}{\pi} \frac{1}{i} \text{Log}_{\mathcal{H}} z_2(\rho).$$

For the midpoint event, $u_\rho = \pi/6$, hence

$$\ell(\rho) = \frac{3}{\pi} \cdot \frac{\pi}{6} = \frac{1}{2},$$

so the projected $1/2$ is the logarithmic readout of the $\pi/3$ unit-circle coordinate. The branch of $\text{Log}_{\mathcal{H}}$ is fixed by the continuous helix lift through the crossing, so the height is retained by the lifted phase rather than by the collapsed radial coordinate.

Definition 7.2 (Fiber capture event). *For a Dirichlet channel χ ,*

$$\text{FiberCaptureEvent}_{\chi}(\rho) \quad :\Longleftrightarrow \quad \rho \in \text{NTZ}(\chi).$$

Definition 7.3 (Source fiber event). *For a non-principal Dirichlet channel χ and point ρ , a source fiber event is a three-dimensional event over ρ carrying the following data:*

(S1) a source drift rate λ_ρ and nonzero amplitude a_ρ with

$$e^{\text{Re}(\lambda_\rho)\tau} a_\rho = a_\rho \quad (\forall \tau \in \mathbb{R});$$

(S2) the $\pi/3$ source coordinate

$$x_\rho = \text{arcChart}(\text{Re } \rho) = \frac{\pi}{3} \text{Re } \rho;$$

(S3) height retention

$$\text{Im}(\lambda_\rho) = \text{Im } \rho;$$

(S4) nonzero spectral value $\rho \neq 0$;

(S5) source-chain cancellation with explicit decay;

(S6) the resolvent-trace pole at ρ .

It does not contain the conclusion $\text{Re } \rho = 1/2$.

Theorem 7.4 (Source completeness from R1–R9). *For every non-principal χ , every fiber capture event $\rho \in \text{NTZ}(\chi)$ is realized by a source fiber event.*

Proof. This is the Lean theorem `sourceComplete3D_of_HP`. It uses: R5 for source-chain cancellation and the resolvent pole, and R9 to identify the Mathlib zero-set object. The produced source event records the R8 coordinate language by setting its source coordinate to $\text{arcChart}(\text{Re } \rho) = (\pi/3) \text{Re } \rho$. The midpoint translation $\text{arcChart}(\text{Re } \rho) = \pi/6 \iff \text{Re } \rho = 1/2$ is used later in `SourceFiberEvent.midpoint_axis_of_HP`. Source completeness itself does not prove $\text{Re } \rho = 1/2$. $\square$

8 Unconditional Shifted-Coordinate Closure

Definition 8.1 (Shifted midpoint readout). *For a captured zero ρ , define the shifted crossing readout*

$$\rho^\sharp := \frac{1}{2} + i \Im \rho.$$

Equivalently, in the Lean file this is

$$\text{shiftedMidpointZero } \rho.$$

The real coordinate of $\rho^\sharp$ is the normalized midpoint readout, and the imaginary coordinate is the retained source height:

$$\text{Re } \rho^\sharp = \frac{1}{2}, \quad \text{Im } \rho^\sharp = \text{Im } \rho.$$

Theorem 8.2 (Per-character shifted-coordinate discharge). *Let $\chi \neq 1$. For every fiber capture event $\rho \in \text{NTZ}(\chi)$, there exists a shifted readout zero $\rho^\sharp$ such that*

$$\text{Re } \rho^\sharp = \frac{1}{2}, \quad \text{Im } \rho^\sharp = \text{Im } \rho.$$

Proof. This is the Lean theorem `shiftedCoordinateDischarge_of_HP`. The proof uses no projection-faithfulness hypothesis.

Let $\rho \in \text{NTZ}(\chi)$. By `sourceComplete3D_of_HP`, R1–R9 realize ρ as a source fiber event. The geometric source projection is

$$u \mapsto u + \left(\frac{\pi}{6} - u \right),$$

formalized as `sourcePiThirdGeometricProjection`. Hence, by `sourcePiThirdGeometricProjection_midpoint`,

$$\text{sourcePiThirdGeometricProjection}(u) = \frac{\pi}{6}.$$

The logarithmic one-dimensional readout in $\pi/3$ units is

$$\text{piThirdLogReadout}(u) = \frac{3}{\pi} u,$$

so the projected midpoint reads

$$\text{piThirdLogReadout}\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

by `piThirdLogReadout_midpoint`. The crossing event is then packaged as `SourceShiftedCrossingEvent`: its readout is $\rho^\sharp = \text{shiftedMidpointZero } \rho$, so

$$\text{Re } \rho^\sharp = \frac{1}{2}, \quad \text{Im } \rho^\sharp = \text{Im } \rho.$$

$\square$

Theorem 8.3 (All non-principal shifted-coordinate discharge). *For every modulus M , every non-principal Dirichlet channel $\chi : \text{DirichletCharacter}(\mathbb{C}, M)$, and every captured zero $\rho \in \text{NTZ}(\chi)$, there exists a shifted readout zero $\rho^\sharp$ such that*

$$\text{Re } \rho^\sharp = \frac{1}{2}, \quad \text{Im } \rho^\sharp = \text{Im } \rho.$$

Proof. Apply the per-character discharge to each M and each $\chi \neq 1$. In Lean this is `GRHComplete_shiftedCoordinateDischarge_of_HP`, with the underlying all-channel statement `shiftedCoordinateDischargeAll_of_HP`. $\square$

Theorem 8.4 (Zeta/L1 shifted-coordinate discharge). *For every zeta/L1 nontrivial zero $\rho \in \text{NTZ}_\zeta$, there exists a shifted readout zero $\rho^\sharp$ such that*

$$\text{Re } \rho^\sharp = \frac{1}{2}, \quad \text{Im } \rho^\sharp = \text{Im } \rho.$$

Proof. This is `RH_shiftedCoordinateDischarge_of_HP`, built from `zetaShiftedCoordinateDischarge_of_HP`. The readout is again $\rho^\sharp = \text{shiftedMidpointZero } \rho$. $\square$

Remark 8.5 (Compatibility wrappers). *The earlier structures `FaithfulDimensionalProjection`, `GRH_of_HP_and_Faithfulness`, `GRHComplete_of_HP_and_Faithfulness`, and `RH_of_HP_and_Faithfulness` remain in the Lean file as compatibility wrappers for the original-coordinate Pythagorean transport formulation. The current unconditional discharge used by the paper is the shifted-coordinate readout above.*

9 Mathematical Reproduction of the Computation

The computation evaluates the source fiber equations above. It is not a proof step. The crossing ordinates are defined by the complete geometric fiber and its standing readout. Finite summation, residue decomposition, Hurwitz notation, and Euler–Maclaurin summation appear only where the derived fiber is represented for one-dimensional evaluation; they are not sources of the fiber.

9.1 Placed Geometric Fiber

The geometry is

$$A = \pi/6, \quad ds = \pi/3, \quad C = 2A ds = (\pi/3)^2.$$

For $s = \frac{1}{2} + it$, the placed finite head is

$$H_M(t) = \sum_{n=1}^M \chi(n) R_n^{-1} e^{-it \text{Log}(R_n^2)}, \quad R_n^2 = Cn.$$

Thus

$$H_M(t) = \sum_{n=1}^M \chi(n) (Cn)^{-s}.$$

The tail is not added from a separate external object. It is the remaining geometric fiber beyond the cutoff:

$$G_M(t) = \sum_{n>M} \chi(n) R_n^{-1} e^{-it \text{Log}(R_n^2)} = \sum_{n>M} \chi(n) (Cn)^{-s}.$$

The complete placed fiber is

$$F_\chi(t) = H_M(t) + G_M(t),$$

independently of the cutoff M when the tail is included.

9.2 Residue Coordinates for the Geometric Tail

The geometric tail may be written in conductor residue coordinates. Let $n_{0,r}$ be the first integer $> M$ in residue class $r \bmod q$. Then

$$G_M(t) = \sum_r \chi(r) C^{-s} q^{-s} \sum_{m \geq 0} \left(\frac{n_{0,r}}{q} + m \right)^{-s}.$$

This identity is just the residue-class reindexing of the geometric terms $R_n^2 = Cn$. Hurwitz notation abbreviates the inner residue tail; it does not define the tail or the crossing condition. It is a coincidental representation of the derived fiber.

For finite one-dimensional evaluation, replace the inner infinite residue tail by

$$\mathcal{T}_{h,J}(s, x) \approx \sum_{m \geq 0} (x + m)^{-s}.$$

For integers $h, J \geq 1$, set $X = x + h$ and write

$$(s)^{\overline{m}} = s(s+1) \cdots (s+m-1), \quad (s)^{\overline{0}} = 1.$$

The Euler–Maclaurin summation evaluator used for that geometric residue tail is

$$\begin{aligned} \mathcal{T}_{h,J}(s, x) &= \sum_{m=0}^{h-1} (x + m)^{-s} + \frac{X^{1-s}}{s-1} + \frac{1}{2} X^{-s} \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{(2j)!} (s)^{\overline{2j-1}} X^{-s-2j+1}, \end{aligned}$$

where B_{2j} are Bernoulli numbers. The associated remainder is the standard Euler–Maclaurin remainder after the B_{2J} term. The numerical tables in the repository use $h = 8$ and finite J chosen large enough for the working precision.

With this evaluator, the finite computed fiber is

$$F_{M,\chi}(t) = \sum_{n=1}^M \chi(n) (Cn)^{-s} + \sum_r \chi(r) C^{-s} q^{-s} \mathcal{T}_{h,J} \left(s, \frac{n_{0,r}}{q} \right).$$

For the L1 channel, the same geometric tail is grouped with conductor 2 and weights

$$\chi_\eta(1) = 1, \quad \chi_\eta(0) = -1,$$

so that

$$F_{M,L1}(t) = C^{-s} 2^{-s} \left[\mathcal{T}_{h,J} \left(s, \frac{n_{0,1}}{2} \right) - \mathcal{T}_{h,J} \left(s, \frac{n_{0,0}}{2} \right) \right] + \sum_{n=1}^M (-1)^{n+1} (Cn)^{-s}.$$

This is the finite evaluation of the eta-regularized zeta fiber in placed geometric coordinates.

9.3 Principal One-Dimensional Tail Error Control

For the projected principal/L1 tail, the repository also formalizes the classical Euler–Maclaurin summation identity as an error-control theorem. Let

$$S_N(s) = \sum_{n=1}^N n^{-s}, \quad \{u\} = u - \lfloor u \rfloor,$$

and define

$$I(s, N) = \int_N^\infty \{u\} u^{-s-1} du, \quad R(s, N) = sI(s, N).$$

For $\operatorname{Re} s > 0$ and $s \neq 1$, the continued one-dimensional response is

$$c(s) = \frac{s}{s-1} - s \int_1^\infty \{u\} u^{-s-1} du,$$

and the Euler–Maclaurin identity is

$$c(s) = \zeta(s),$$

with finite-readout error

$$S_N(s) - \zeta(s) - \frac{N^{1-s}}{1-s} = R(s, N)$$

and bound

$$\|R(s, N)\| \leq \frac{\|s\|}{\operatorname{Re} s} N^{-\operatorname{Re} s} \quad (N \geq 2).$$

This theorem controls the finite projected evaluation of the principal geometric tail; it is a representation of the derived fiber, not a source for the tail.

The full finite readout is

$$Z_\chi(t) = \operatorname{Re}(e^{i\Theta_\chi(t)} F_{M,\chi}(t)).$$

Crossings are isolated by sign changes

$$Z_\chi(a)Z_\chi(b) < 0$$

followed by bisection. If $\gamma \in (a, b)$ is a simple crossing,

$$Z_\chi(\gamma) = 0, \quad Z'_\chi(\gamma) \neq 0,$$

then there is an interval around γ on which the sign changes once, and bisection on any bracket inside that interval converges to γ .

For local reproduction from an intermediate height t_0 , choose $\delta > 0$ and evaluate Z_χ on $[t_0 - \delta, t_0 + \delta]$. If a sign change occurs in $[a, b] \subset [t_0 - \delta, t_0 + \delta]$, bisection on Z_χ rederives the crossing in that local interval; no earlier crossing is used.

10 Formal and Computational Support

The proof above is the mathematical statement. The repository support has two roles: the formal files check the stated implications in Lean, and the numerical files evaluate the displayed fiber equations.

10.1 Formalization Map

Dirichlet zero set	GRHSpectral.NontrivialZeros
per-character GRH predicate	GRHSpectral.GRH
zeta zero set	ZD.NontrivialZeros
Euler–Maclaurin continuation	GRH.EulerMaclaurinDirichlet.c.fun
Euler–Maclaurin error term	GRH.EulerMaclaurinDirichlet.R
Euler–Maclaurin identity	GRH.EulerMaclaurinDirichlet.difference.eq_R
Euler–Maclaurin bound	GRH.EulerMaclaurinDirichlet.euler_maclaurin_dirichlet_explicit
zeta one-dimensional completion	SourceContinuation.zetaCompletion
residue fiber decomposition	DirichletFibre.LFunction.eq_fibre_sum
fiber cancellation	DirichletFibre.zero_iff_fibre_cancellation
source no-drift	HelixSource.source_noDrift
helix continuation into strip	helixOwnContinuationIntoStrip
flow continuation estimate	HelixFlowClosureLedger.flow_rate_bound
zero cancellation estimate	HelixFlowClosureLedger.flow_chain_vanishes_at_zero
height-free source ledger	HelixFlowClosureLedger.flow_closure_exact
FTA angle additivity	HelixLogFree.windAngle_mul
FTA winding multiplicativity	HelixLogFree.wind_mul
channel helix multiplicativity	HelixMult.helixChar_mul
log bridge phasor	HelixWindBridge.wind_glog.eq_cpow
helix Euler product	HelixMult.helixSource.eq_eulerProduct
winded prime trace shift	HelixWindBridge.windedVonMangoldt.eq_neg_logDeriv_shift
Euler-product edge nonvanishing	HelixMult.helix_no_zero_re_ge.one
Euler-product discriminator	EulerProductDiscriminator
Dirichlet discriminator constructor	eulerProductDiscriminatorOfDirichlet
prime flow trace	HelixFlowVonMangoldt.flowVonMangoldtTrace.eq_negLogDeriv
exact multiplicity capture	HelixLimit.multiplicityCapture_of_resolventTrace
Möbius unit readout	moebius_unit_iff
Pythagorean readout	readout_pythagoras_iff_online
spectral unit midpoint	SpectralSide.w_unit_iff_half
lifted crossing circle value	liftedCrossingCircleValue
lifted logarithmic readout	liftedCrossingLog
$\pi/3$ log-readout midpoint	piThirdLogReadout_midpoint
$\pi/3$ gauge	HelixGauge.piThirdGauge.eq
nonprincipal closed form	HelixGauge.piThirdHelixSource.eq_gauge_mul.L
geometric trace closed form	HelixGauge.helixTrace.eq_gauge_mul_negLogDeriv
$\pi/3$ continued trace	HelixGauge.piThirdHelixTraceCont.eq
completed trace gauge split	completedLFunction.logDeriv_gauge_split
prime/dual trace meeting	flowDualTraceMeeting
strip trace continuation	primeTraceContinuationIntoStrip
L1 closed form	HelixGauge.piThirdZetaFiber
L1 zero equivalence	HelixGauge.piThirdZetaFiber_zero_iff
L1 gauged continued trace	HelixGauge.piThirdZetaTraceCont.eq
L1 resolvent pole	HelixGauge.piThirdZetaFiber_zero_resolventTrace_pole
L1 gauged resolvent pole	HelixGauge.piThirdZetaFiber_zero_zetaTraceCont_pole
L1 standing-wave node readout	HelixStandingWave.zeta_zero_on_line_iff_standingWave_node
L1 helix fiber standing node	zetaL1HelixFiber_zero_iff_standingWave_node
L1 Euler-product discriminator	ZetaEulerProductDiscriminator
L1 discriminator constructor	zetaEulerProductDiscriminator
R1–R9 infrastructure	hilbertPolyaChainR1ToR9
source fiber event	SourceFiberEvent
source completeness predicate	SourceComplete3D
source completeness from R1–R9	sourceComplete3D_of_HP
geometric midpoint projection	sourcePiThirdGeometricProjection_midpoint
shifted midpoint zero	shiftedMidpointZero
per-character shifted discharge	shiftedCoordinateDischarge_of_HP
all-channel shifted discharge	GRHComplete.shiftedCoordinateDischarge_of_HP
zeta/L1 shifted discharge	RH.shiftedCoordinateDischarge_of_HP

10.2 Lean Verification Protocol

File-level verification:

```
lake env lean RequestProject/HilbertPolyaChain.lean
lake env lean RequestProject/HelixLogFreeFTA.lean
lake env lean RequestProject/HelixMultiplicative.lean
```

```
lake env lean RequestProject/HelixWindBridge.lean
lake env lean RequestProject/HelixStandingWave.lean
```

Closed-form gauge verification:

```
lake env lean RequestProject/HelixGaugeIdentity.lean
```

Escape-hatch audit:

```
rg -n "axiom |sorry|admit|@[implemented_by\]" \
  RequestProject/HilbertPolyaChain.lean
rg -n "axiom |sorry|admit|@[implemented_by\]" \
  RequestProject/HelixLogFreeFTA.lean \
  RequestProject/HelixMultiplicative.lean \
  RequestProject/HelixWindBridge.lean
```

Axiom footprint targets:

```
#print axioms hilbertPolyaChainR1ToR9
#print axioms helixOwnContinuationIntoStrip
#print axioms zetaL1HelixFiber_zero_iff_standingWave_node
#print axioms HelixFlowClosureLedger.flow_rate_bound
#print axioms HelixFlowClosureLedger.flow_chain_vanishes_at_zero
#print axioms HelixFlowClosureLedger.flow_closure_exact
#print axioms HelixLogFree.windAngle_mul
#print axioms HelixLogFree.wind_mul
#print axioms HelixMult.helixChar_mul
#print axioms HelixWindBridge.real_log_eq_factorization_sum
#print axioms HelixWindBridge.wind_glog_eq_cpow
#print axioms HelixMult.helixSource_eq_eulerProduct
#print axioms HelixWindBridge.windedVonMangoldt_eq_neg_logDeriv_shift
#print axioms HelixMult.helix_no_zero_re_ge_one
#print axioms eulerProductDiscriminatorOfDirichlet
#print axioms zetaEulerProductDiscriminator
#print axioms HelixFlowVonMangoldt.flowVonMangoldtTrace_eq_negLogDeriv
#print axioms HelixLimit.multiplicityCapture_of_resolventTrace
#print axioms HelixGauge.helixTrace_eq_gauge_mul_negLogDeriv
#print axioms completedLFunction_logDeriv_gauge_split
#print axioms flowDualTraceMeeting
#print axioms primeTraceContinuationIntoStrip
#print axioms sourceComplete3D_of_HP
#print axioms sourcePiThirdGeometricProjection_midpoint
#print axioms shiftedMidpointZero_re
#print axioms shiftedMidpointZero_im
#print axioms sourceShiftedCrossingEvent_of_HP
#print axioms shiftedCoordinateDischarge_of_HP
#print axioms shiftedCoordinateDischargeAll_of_HP
#print axioms GRHComplete_shiftedCoordinateDischarge_of_HP
#print axioms piThirdLogReadout_midpoint
#print axioms zetaShiftedCoordinateDischarge_of_HP
```



```

#print axioms RH_shiftedCoordinateDischarge_of_HP
#print axioms HelixGauge.piThirdGauge_eq
#print axioms HelixGauge.piThirdZetaFiber_zero_iff
#print axioms HelixGauge.piThirdZetaTraceCont_eq
#print axioms HelixGauge.piThirdZetaFiber_zero_resolventTrace_pole
#print axioms HelixGauge.piThirdZetaFiber_zero_zetaTraceCont_pole
#print axioms HelixStandingWave.zeta_zero_on_line_iff_standingWave_node

```

Expected footprint:

[propext, Classical.choice, Quot.sound].

11 Current Logical Status

HP R1–R9 infrastructure	Lean theorem	hilbertPolyaChainR1ToR9
Helix continuation into strip	Lean theorem	helixOwnContinuationIntoStrip
L1 helix standing readout	Lean theorem	zetaL1HelixFiber_zero_iff_ standingWave_node
$\pi/3$ closed form	Lean theorems	HelixGauge.piThird*
Euler-product discriminator	Lean definitions	EulerProductDiscriminator / ZetaEulerProductDiscriminator
Source completeness	Lean theorem	sourceComplete3D_of_HP
Geometric midpoint projection	Lean theorem	sourcePiThirdGeometricProjection_ midpoint
Shifted readout event	Lean structure	SourceShiftedCrossingEvent
Per-character shifted discharge	Lean theorem	shiftedCoordinateDischarge_of_HP
All non-principal shifted discharge	Lean theorem	GRHComplete_ shiftedCoordinateDischarge_of_HP
Zeta/L1 shifted discharge	Lean theorem	RH_shiftedCoordinateDischarge_ of_HP

The older original-coordinate projection transport wrappers are still present for compatibility. Their atoms are the following fields:

source_event_faithfully_projected	$\text{SourceFiberEvent}_\chi(\rho) \Rightarrow \text{Proj}_\chi(\rho)$
lower_dimensions_create_no_events	$\text{Proj}_\chi(\rho) \Rightarrow \text{SourceFiberEvent}_\chi(\rho)$
projected_readout_pythagorean	$\mathcal{E}_\chi \Rightarrow \text{Proj}_\chi(\rho) \Rightarrow \ w(\rho)\ ^2 = 1$

The unconditional shifted-coordinate discharge does not consume these fields. It consumes source completeness, the geometric midpoint projection in $\pi/3$ coordinates, and the retained-height definition of `shiftedMidpointZero`.